

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
 - Write all answers sequentially.
 - Start answering a question (not the part of the question) from the top of a new page.
 - **Write legibly and in orderly fashion maintaining all mathematical norms and rules.**
 - Start working right away. There is no late submission form. If you miss the deadline, you need to use the make-up assignment to cover up the marks.
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1. A linear system is described by the following equations:

$$x_1 + 6x_2 + 2x_3 = 10$$

$$3x_1 + 2x_2 + x_3 = 6$$

$$4x_1 + 5x_2 + 2x_3 = 9.$$

Based on these equations, answer the questions below.

- [1 marks] From the given linear equations, identify the matrices A , x and b such that the linear system can be expressed as a matrix equation.
- [2 marks] Construct the Frobenius matrices $F^{(1)}$ and $F^{(2)}$ from this system.
- [1 marks] Compute the unit lower triangular matrix L .
- [3 marks] Now find the solution of the linear system using LU decomposition method. Use the unit lower triangular matrix found in the previous question.

2. A linear system is described by the following equations:

$$6x_2 + 2x_3 = 10$$

$$3x_1 + 2x_2 + x_3 = 6$$

$$4x_1 + 5x_2 + 2x_3 = 9.$$

Based on these equations, answer the questions below.

- [1 marks] From the given linear equations, identify the matrices A , x and b such that the linear system can be expressed as a matrix equation.
- [1 marks] Examine if the matrix A has any pivoting problem? Explain why or why not?
- [3 marks] Write down the Augmented matrix, $\text{Aug}(A)$, from the given linear system, and evaluate the upper triangular matrix U . Note that you have to show the row multipliers m_{ij} for each step as necessary.
- [3 marks] Using the upper triangular matrix found in the previous question, compute the solution of the given linear system by Gaussian elimination method.